Chapter 7

Build iOS Bridge Components with Swift

You just learned how to build fully native screens with SwiftUI and Jetpack Compose. And as I'm sure you noticed, there's a *lot* of code needed to get it all working. Not to mention coordinating deploys between your Rails APIs and the apps in the app stores.

So what's a savvy hybrid app developer like yourself to do if you only need a native component or two? Say you want to level up your sign-in screen with a native button in the navigation bar at the top. Enter bridge components. They unlock *progressive enhancement* of individual controls without converting entire screens to native. Bridge components strike a nice balance between HTML screens and fully native ones. They require less upfront code to get working and, when built correctly, can be sprinkled onto new screens without needing native code updates. This means less code to write and less code to maintain.

Here are two examples from the hotwire-native-ios demo app.[37] For each, the web-based HTML is on the left and the bridge component is on the right.

The first example is a simple form, as shown in the first set of screenshots.